

```
In [ ]: List is a collection items.  
We can store heterogeneous data types.  
int, float, complex,list,dict,str,  
set, tuple, pandas series,pandas dataframe etc.
```

Properties:

```
In [ ]: 1)List is a mutable>> changeable(Add, update ,delete)  
2)It is ordered data types.  
3)Duplicate items are allowed.  
4)List items enclosed by [brackets]  
5)List items are comma separated.
```

```
In [ ]: List Initialization  
[] or list()
```

```
In [2]: l1=list()  
l1
```

```
Out[2]: []
```

```
In [1]: l2=[]  
l2
```

```
Out[1]: []
```

List Functions

```
In [ ]: 1)append()  
2)extend()  
3)insert()  
4)remove()  
5)pop()  
6)clear()
```

List Indexing

```
In [ ]: Indexing always start from 0(zero).
```

```
In [3]: list1=[4,54,3,46,67,7,8]  
list1
```

```
Out[3]: [4, 54, 3, 46, 67, 7, 8]
```

```
In [4]: len(list1)
```

```
Out[4]: 7
```

```
In [5]: list1=[4,54,3,46,67,7,8]
print(list1[0])
print(list1[1])
print(list1[2])
print(list1[3])
```

```
4
54
3
46
```

```
In [6]: list1=[4,54,3,46,67,7,8]
print(list1[-1])
print(list1[-2])
print(list1[-3])
print(list1[-4])
```

```
8
7
67
46
```

```
In [ ]: list1= [4,    54,    3,    46,    67,    7,    8]
+ve  =  0      1      2      3      4      5      6
-ve   -7     -6     -5     -4     -3     -2     -1
```

```
In [7]: print(list1[4])
print(list1[-3])
```

```
67
67
```

Slicing

```
In [ ]: We can access one or more items.
```

```
In [ ]: list[start_index,end_index,[step_size]]
```

```
start_index=0
end_index=len(list1)
step_size=1
```

```
start_index=included
end_index=excluded
```

```
In [8]: list1=[4,6,7,1,3,9,2]
        list1[1:5]#6,7,1,3
```

```
Out[8]: [6, 7, 1, 3]
```

```
In [9]: list1=[4,6,7,1,3,9,2]
        list1[3:6]
```

```
Out[9]: [1, 3, 9]
```

```
In [ ]: list1= [4,      6,   7,      1,   3,      9,      2]
+ve    =  0       1   2       3       4       5       6
-ve     -7      -6  -5      -4      -3      -2      -1
```

```
In [10]: list1=[4,6,7,1,3,9,2]
         list1[-6:-2]#6,7,1,3
```

```
Out[10]: [6, 7, 1, 3]
```

```
In [11]: list1=[4,6,7,1,3,9,2]
         list1[-5:-2]
```

```
Out[11]: [7, 1, 3]
```

```
In [12]: list1=[4,6,7,1,3,9,2]
         list1[0:3]
```

```
Out[12]: [4, 6, 7]
```

```
In [13]: list1=[4,6,7,1,3,9,2]
         list1[:3]
```

```
Out[13]: [4, 6, 7]
```

```
In [14]: list1=[4,6,7,1,3,9,2]
         list1[1:]
```

```
Out[14]: [6, 7, 1, 3, 9, 2]
```

```
In [15]: list1=[4,6,7,1,3,9,2]
         list1[-4:]
```

```
Out[15]: [1, 3, 9, 2]
```

```
In [ ]: list1= [4,      6,   7,      1,   3,      9,      2]
+ve    =  0       1   2       3       4       5       6
-ve     -7      -6  -5      -4      -3      -2      -1
```

```
In [16]: list1=[4,6,7,1,3,9,2]
print(list1[2:6])#7,1,3,9
print(list1[-5:-1])#7,1,3,9
print(list1[2:-1])#7,1,3,9
print(list1[-5:6])#7,1,3,9
```

```
[7, 1, 3, 9]
[7, 1, 3, 9]
[7, 1, 3, 9]
[7, 1, 3, 9]
```

```
In [17]: list1=[4,6,7,1,3,9,2]
list1[:]
```

```
Out[17]: [4, 6, 7, 1, 3, 9, 2]
```

```
In [18]: list1=[4,6,7,1,3,9,2]
list1[0:len(list1)]
```

```
Out[18]: [4, 6, 7, 1, 3, 9, 2]
```

```
In [19]: list1=[4,6,7,1,3,9,2]
list1[::]
```

```
Out[19]: [4, 6, 7, 1, 3, 9, 2]
```

Reverse List

```
In [20]: list1=[4,6,7,1,3,9,2]
list1[::-1]
```

```
Out[20]: [2, 9, 3, 1, 7, 6, 4]
```

```
In [21]: list1=[4,6,7,1,3,9,2]
list1[::-2]
```

```
Out[21]: [4, 7, 3, 2]
```

```
In [22]: list1=[4,6,7,1,3,9,2]
list1[5:2]
```

```
Out[22]: []
```

```
In [23]: list1=[4,6,7,1,3,9,2]
list1[1:5:2]
```

```
Out[23]: [6, 1]
```

```
In [ ]: list1= [4,      6,   7,    1,   3,    9,    2]
+ve   =  0      1    2    3    4    5    6
-ve    -7     -6   -5   -4   -3   -2   -1
```

```
In [24]: list1=[4,6,7,1,3,9,2]
list1[1:5:-2]
#your task
```

Out[24]: []

```
In [25]: start_index=int(input("Enter value for start_index:"))
end_index=int(input("Enter value for end_index:-"))
step_size=int(input("Enter value for step size:"))
list1=[4,6,7,1,3,9,2]
list1[start_index:end_index:step_size]
```

```
Enter value for start_index:2
Enter value for end_index:-5
Enter value for step size:1
```

Out[25]: [7, 1, 3]

Access List items by using For Loop

```
In [26]: list1=[4,6,7,10,3,9,2]
len(list1)
```

Out[26]: 7

```
In [27]: list1=[4,6,7,10,3,9,2]
for value in list1:
    print(value)
```

```
4
6
7
10
3
9
2
```

```
In [28]: list1=[4,6,7,10,3,9,2]
for value in list1:
    print("We are in for loop")
```

```
We are in for loop
We are in for loop
We are in for loop
We are in for loop
We are in for loop
We are in for loop
We are in for loop
```

```
In [29]: list1=[4,6,7,10,3,9,2]
for value in list1:
    print("We are in for loop")
    print("value is:",value)
    print("***50")
```

```
We are in for loop
value is: 4
*****

We are in for loop
value is: 6
*****

We are in for loop
value is: 7
*****

We are in for loop
value is: 10
*****

We are in for loop
value is: 3
*****

We are in for loop
value is: 9
*****

We are in for loop
value is: 2
*****
```

```
In [30]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]
for city in city_names:
    print("City name is:",city)
```

```
City name is: Pune
City name is: Mumbai
City name is: Delhi
City name is: Chennai
City name is: Banglore
```

```
In [33]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]
count=0
for city in city_names:
    print(f"City at index {count} is:{city}")
    count+=1
```

```
City at index 0 is:Pune
City at index 1 is:Mumbai
City at index 2 is:Delhi
City at index 3 is:Chennai
City at index 4 is:Banglore
```

```
In [35]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]

for index,city in enumerate(city_names):
    print(index,city)
```

```
0 Pune
1 Mumbai
2 Delhi
3 Chennai
4 Banglore
```

```
In [37]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]

for index,city in enumerate(city_names):
    print(f"City at index {index} is:{city}")
```

```
City at index 0 is:Pune
City at index 1 is:Mumbai
City at index 2 is:Delhi
City at index 3 is:Chennai
City at index 4 is:Banglore
```

```
In [38]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]
city_names[-1]
```

Out[38]: 'Banglore'

```
In [39]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]
city_names[-1]="Hydrabad"
city_names
```

Out[39]: ['Pune', 'Mumbai', 'Delhi', 'Chennai', 'Hydrabad']

```
In [40]: a,b=10,20
a,b=b,a
print(a)
print(b)
```

```
20
10
```

```
In [42]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]
city_names[0],city_names[-1]="Banglore","Pune"
city_names
```

Out[42]: ['Banglore', 'Mumbai', 'Delhi', 'Chennai', 'Pune']

List Functions:

Add Items in List

```
In [ ]: 1)append()  
        2)extend()  
        3)insert()
```

1.append():

```
In [ ]: It is used to add items at last index.  
        Syntax: List.append(value)
```

```
In [43]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]  
        city_names.append("Hydrabad")  
        city_names
```

```
Out[43]: ['Pune', 'Mumbai', 'Delhi', 'Chennai', 'Banglore', 'Hydrabad']
```

```
In [44]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]  
        city_names.append(1000)  
        city_names
```

```
Out[44]: ['Pune', 'Mumbai', 'Delhi', 'Chennai', 'Banglore', 1000]
```

```
In [45]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]  
        city_names.append(5+3j)  
        city_names
```

```
Out[45]: ['Pune', 'Mumbai', 'Delhi', 'Chennai', 'Banglore', (5+3j)]
```

```
In [46]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]  
        city_names.append([3,4,5,6])  
        city_names
```

```
Out[46]: ['Pune', 'Mumbai', 'Delhi', 'Chennai', 'Banglore', [3, 4, 5, 6]]
```

```
In [47]: num=[10,20,30,40,50]  
        num.append(3000)  
        num
```

```
Out[47]: [10, 20, 30, 40, 50, 3000]
```

```
In [48]: city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]  
        num.append(300,400)  
        num
```

TypeError

Traceback (most recent call last)

Cell In[48], line 2

```
1 city_names=["Pune","Mumbai","Delhi","Chennai","Banglore"]  
----> 2 num.append(300,400)  
      3 num
```

TypeError: list.append() takes exactly one argument (2 given)

In []: